```python
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_csv("CustomerChurn.csv")
df.head()
```

```
   customerID  gender  SeniorCitizen Partner Dependents  tenure
PhoneService  \
0  7590-VHVEG  Female              0     Yes         No       1
No
1  5575-GNVDE    Male              0      No         No      34
Yes
2  3668-QPYBK    Male              0      No         No       2
Yes
3  7795-CFOCW    Male              0      No         No      45
No
4  9237-HQITU  Female              0      No         No       2
Yes

      MultipleLines InternetService OnlineSecurity  ...
DeviceProtection  \
0  No phone service             DSL             No  ...
No
1                No             DSL            Yes  ...
Yes
2                No             DSL            Yes  ...
No
3  No phone service             DSL            Yes  ...
Yes
4                No     Fiber optic             No  ...
No

  TechSupport StreamingTV StreamingMovies        Contract
PaperlessBilling  \
0          No          No              No  Month-to-month
Yes
1          No          No              No        One year
No
2          No          No              No  Month-to-month
Yes
3         Yes          No              No        One year
No
4          No          No              No  Month-to-month
Yes

               PaymentMethod MonthlyCharges  TotalCharges Churn
0           Electronic check          29.85         29.85    No
1               Mailed check          56.95        1889.5    No
```

```
2                Mailed check          53.85        108.15   Yes
3  Bank transfer (automatic)          42.30       1840.75    No
4            Electronic check         70.70        151.65   Yes

[5 rows x 21 columns]
```

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   str
 1   gender            7043 non-null   str
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   str
 4   Dependents        7043 non-null   str
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   str
 7   MultipleLines     7043 non-null   str
 8   InternetService   7043 non-null   str
 9   OnlineSecurity    7043 non-null   str
 10  OnlineBackup      7043 non-null   str
 11  DeviceProtection  7043 non-null   str
 12  TechSupport       7043 non-null   str
 13  StreamingTV       7043 non-null   str
 14  StreamingMovies   7043 non-null   str
 15  Contract          7043 non-null   str
 16  PaperlessBilling  7043 non-null   str
 17  PaymentMethod     7043 non-null   str
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   str
 20  Churn             7043 non-null   str
dtypes: float64(1), int64(2), str(18)
memory usage: 1.1 MB
```

Replacing the null value of the totalCharges into 0 and changing the type from obj to float

```
df["TotalCharges"]=df["TotalCharges"].replace(" ","0")

df["TotalCharges"]=df["TotalCharges"].astype(float)

df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
```

```
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   str
 1   gender            7043 non-null   str
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   str
 4   Dependents        7043 non-null   str
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   str
 7   MultipleLines     7043 non-null   str
 8   InternetService   7043 non-null   str
 9   OnlineSecurity    7043 non-null   str
 10  OnlineBackup      7043 non-null   str
 11  DeviceProtection  7043 non-null   str
 12  TechSupport       7043 non-null   str
 13  StreamingTV       7043 non-null   str
 14  StreamingMovies   7043 non-null   str
 15  Contract          7043 non-null   str
 16  PaperlessBilling  7043 non-null   str
 17  PaymentMethod     7043 non-null   str
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   float64
 20  Churn             7043 non-null   str
dtypes: float64(2), int64(2), str(17)
memory usage: 1.1 MB
```

```
df.isnull().sum()
```

```
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
```

```
Churn                0
dtype: int64
```

```
df.describe()
```

```
       SeniorCitizen       tenure  MonthlyCharges  TotalCharges
count    7043.000000  7043.000000     7043.000000   7043.000000
mean        0.162147    32.371149       64.761692   2279.734304
std         0.368612    24.559481       30.090047   2266.794470
min         0.000000     0.000000       18.250000      0.000000
25%         0.000000     9.000000       35.500000    398.550000
50%         0.000000    29.000000       70.350000   1394.550000
75%         0.000000    55.000000       89.850000   3786.600000
max         1.000000    72.000000      118.750000   8684.800000
```

```
df["gender"].value_counts()
```

```
gender
Male      3555
Female    3488
Name: count, dtype: int64
```

checking duplicate values

```
df.duplicated().sum()
```

```
np.int64(0)
```

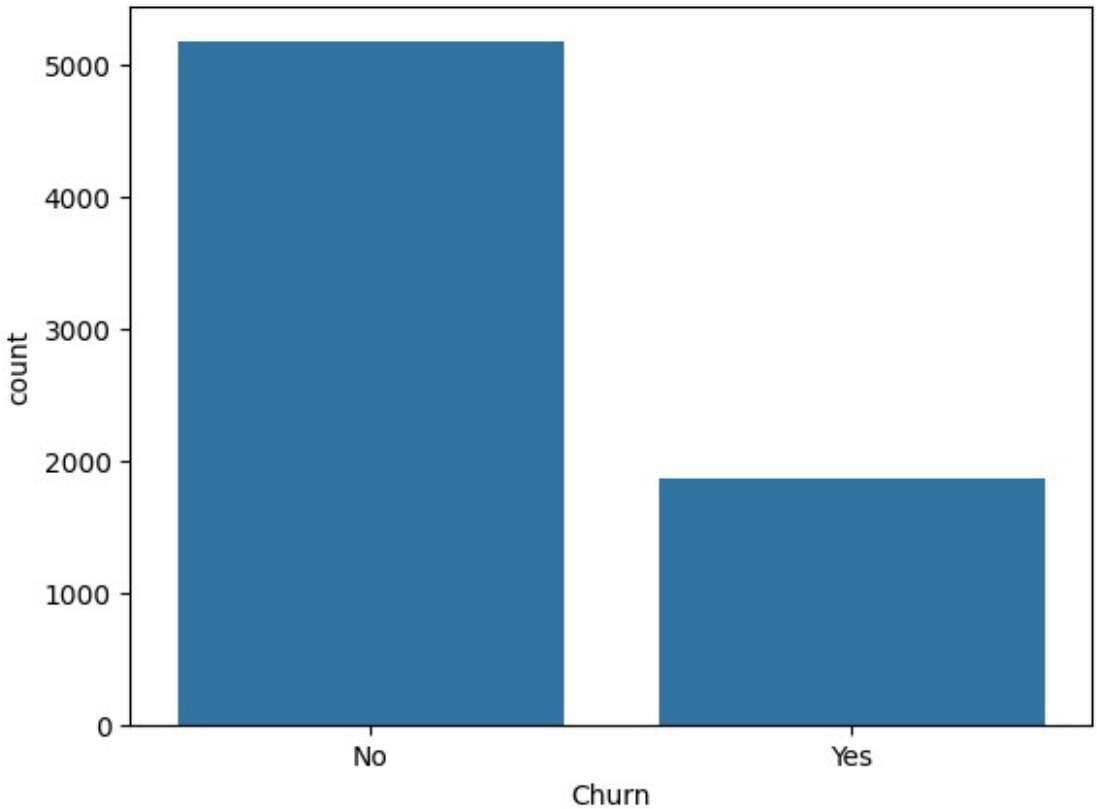
```
df["customerID"].duplicated().sum()
```

```
np.int64(0)
```

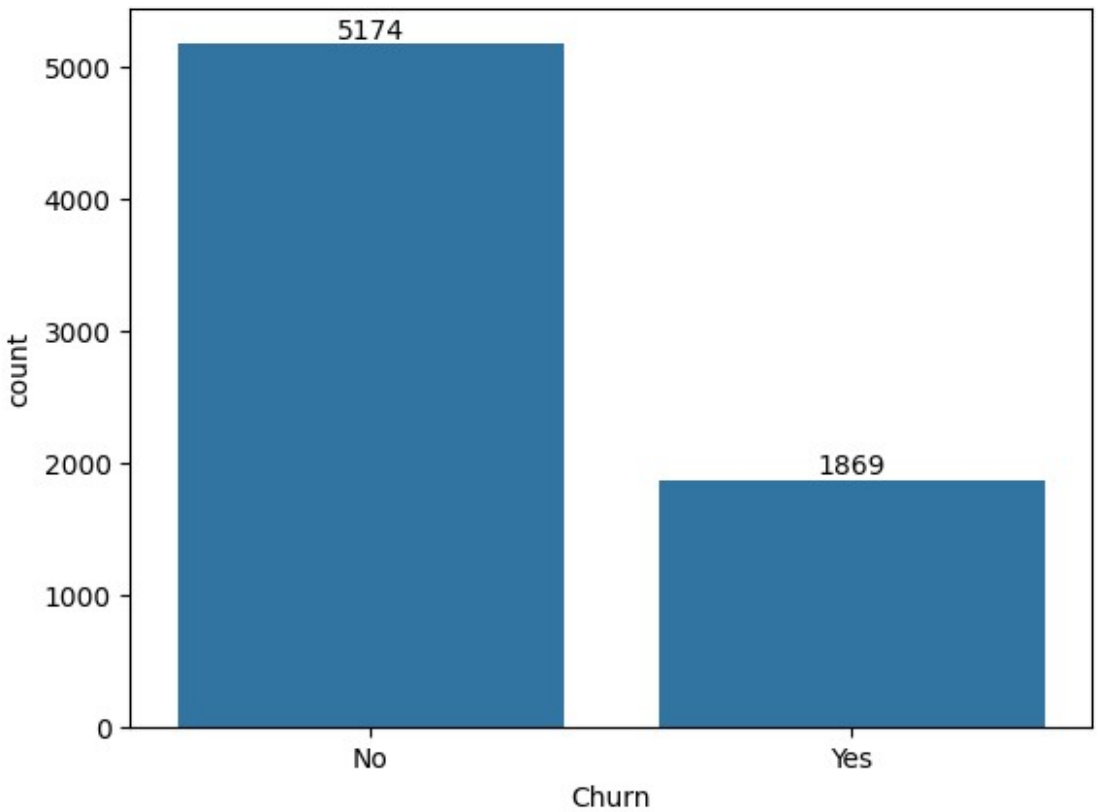
replacing the 1 and 0 to yes and no in the senior citizen list

```
# We can actually do it using replace but let try something new

def conv(i):
    if i==1:
        return "yes"
    else :
        return "no"


df["SeniorCitizen"]=df["SeniorCitizen"].apply(conv)
```

```
sns.countplot(data=df, x="Churn")
plt.show()
```



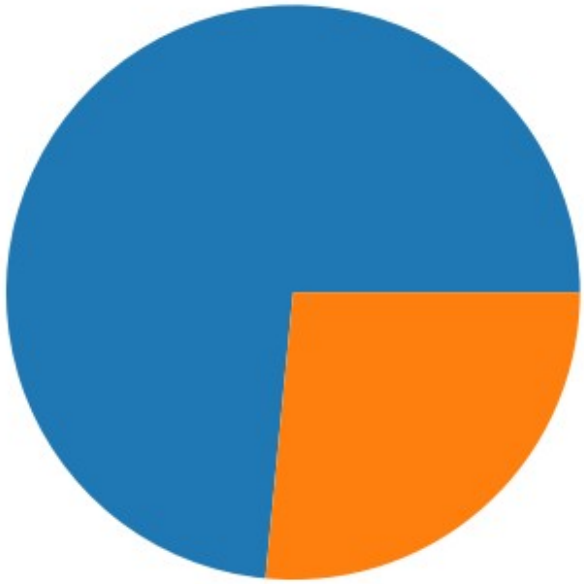

```
ax=sns.countplot(data=df, x="Churn")
ax.bar_label(ax.containers[0])
plt.show()
```



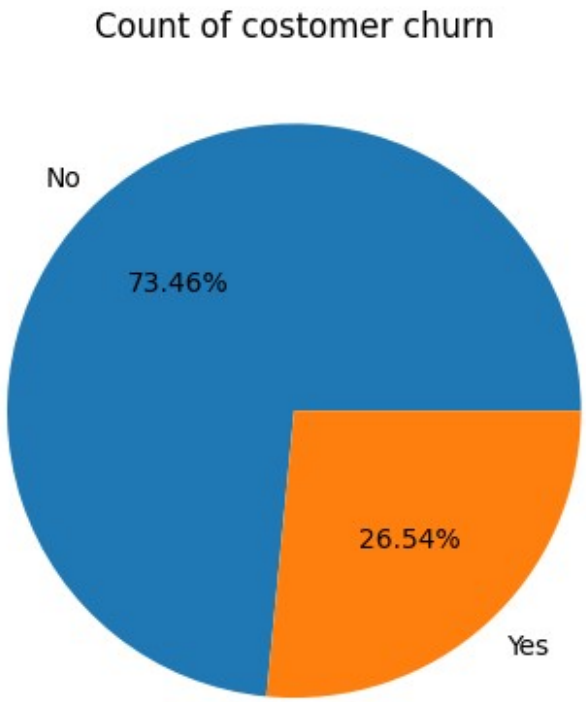

```
# Group by to using pi chart

gb=df.groupby('Churn').agg({"Churn":"count"})
gb
```

```
       Churn
Churn
No      5174
Yes     1869
```

```
plt.pie(x=gb["Churn"])
plt.show()
```



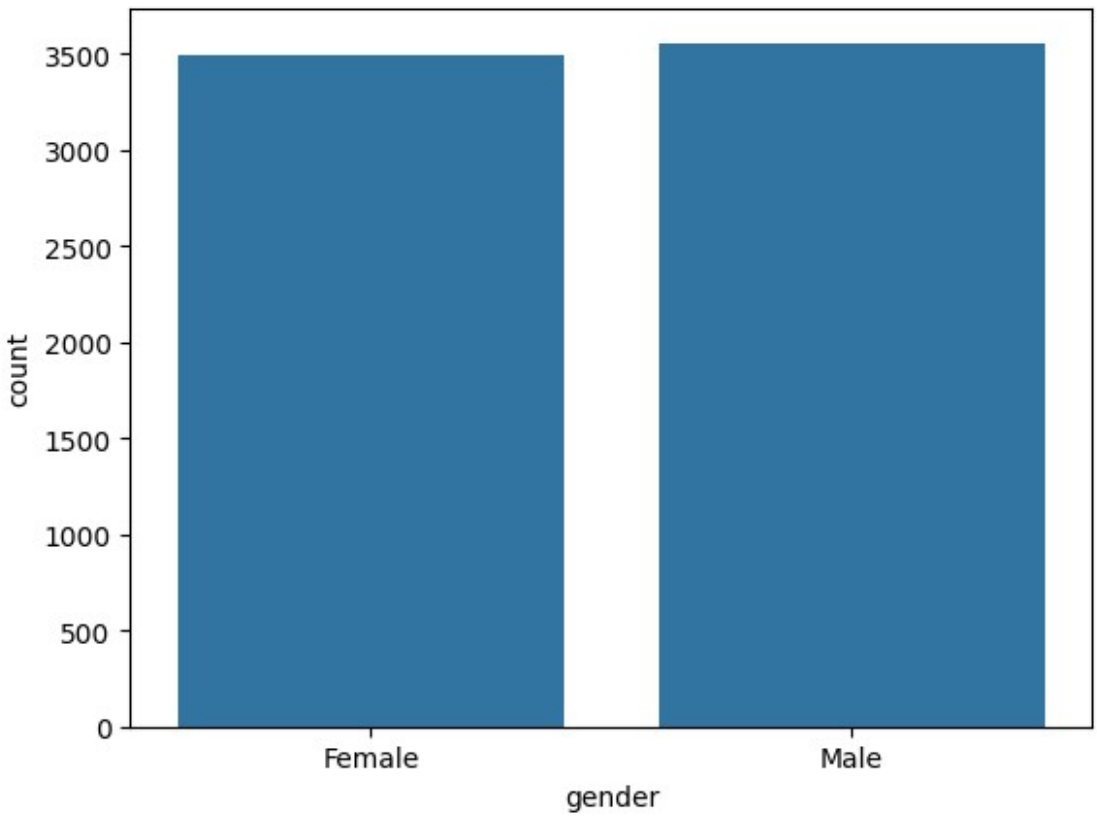
```
x=plt.pie(x=gb["Churn"],labels=gb.index,autopct='%1.2f%%')
plt.title("Count of costomer churn")
plt.show()
```



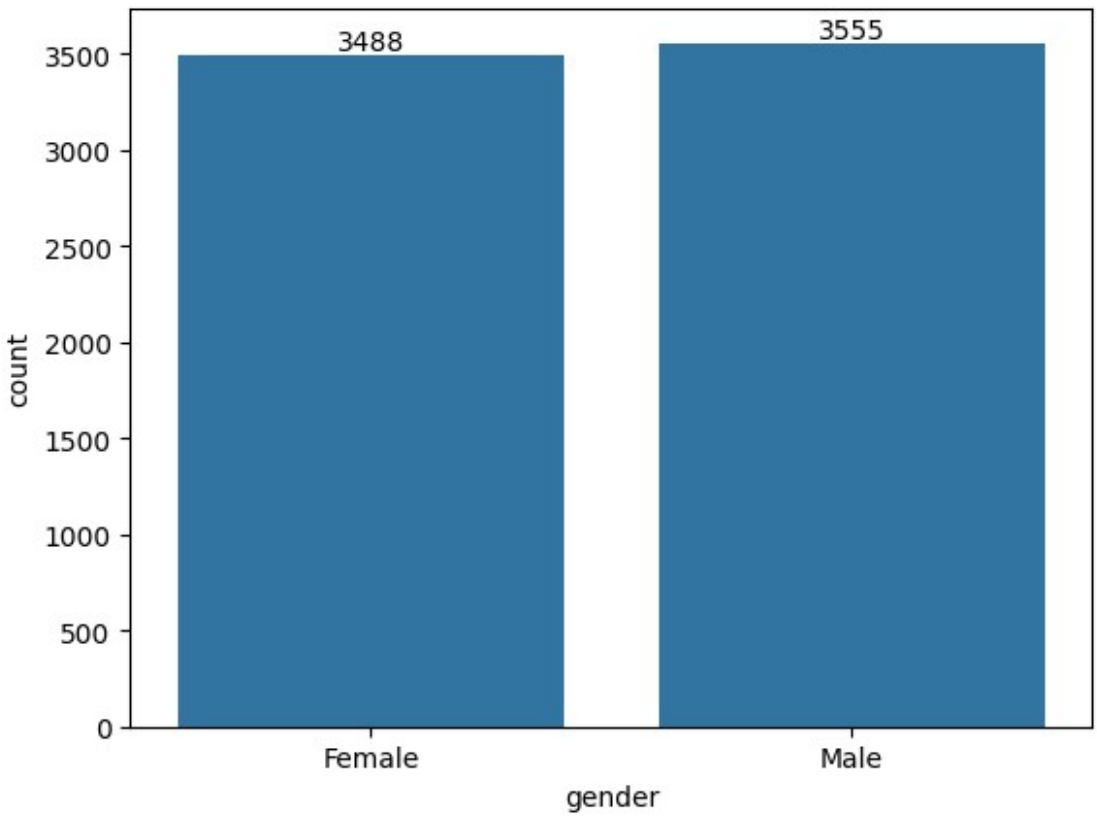
we can conclude that 26.54% costomer are churned out. Now let's explore the reasion behind it

gender

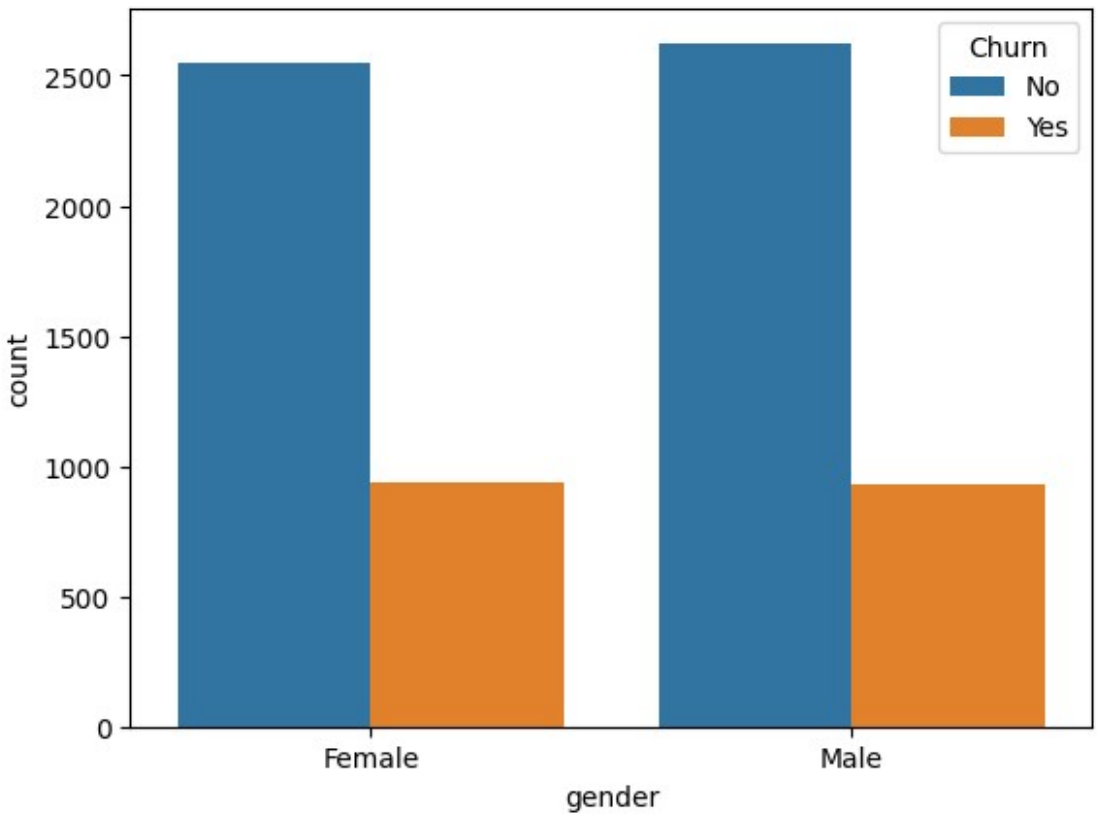
```
sns.countplot(data=df, x='gender')
plt.show()
```



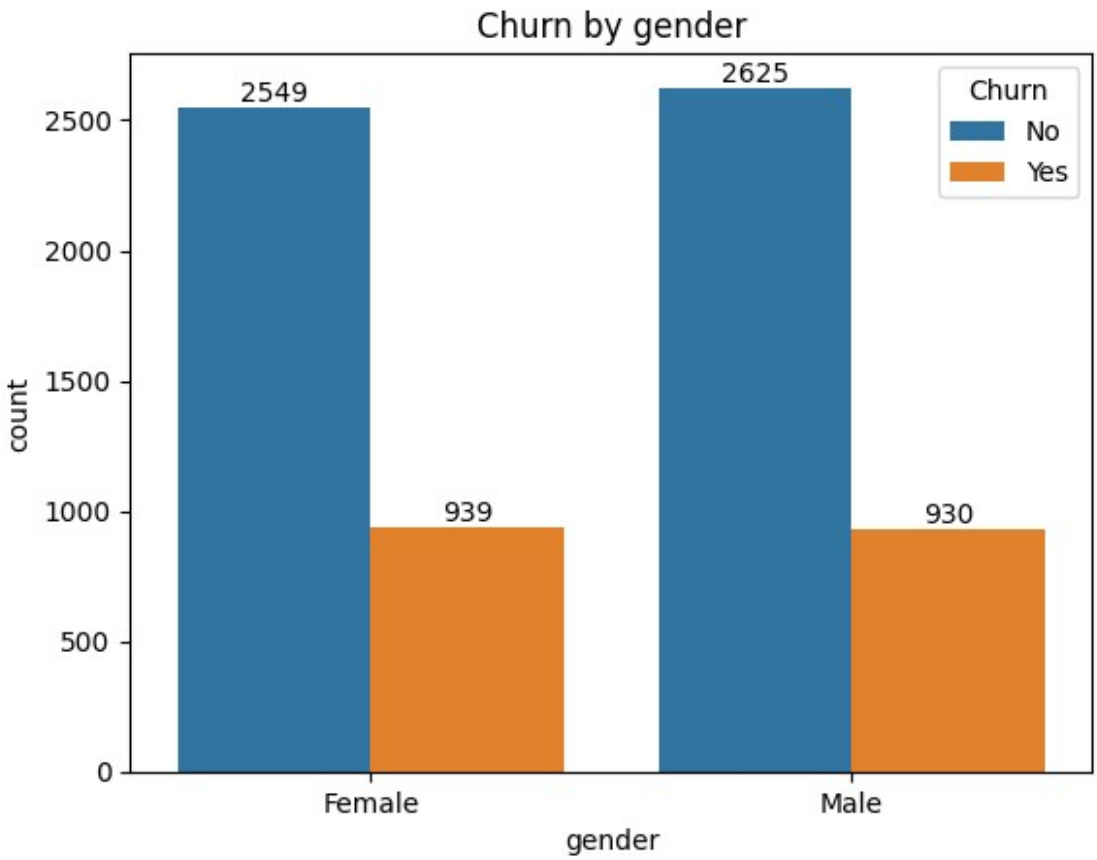

```
ax=sns.countplot(data=df, x='gender')
ax.bar_label(ax.containers[0])
plt.show()
```




```
ax=sns.countplot(data=df, x='gender',hue='Churn')
```



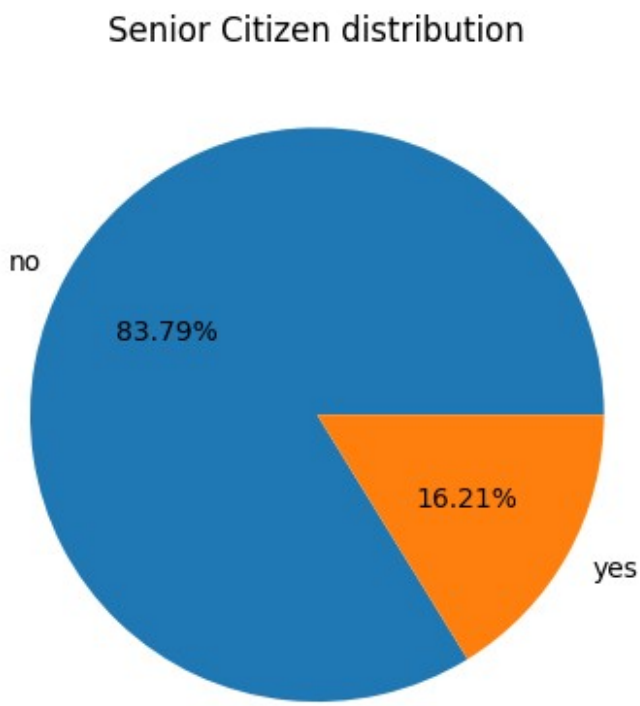

```
ax=sns.countplot(data=df, x='gender',hue='Churn')
for i in ax.containers:
    ax.bar_label(i)
plt.title("Churn by gender")
plt.show()
```




```
gb=df.groupby('SeniorCitizen').agg({'SeniorCitizen':'count'})
gb
```

```
               SeniorCitizen
SeniorCitizen
no                      5901
yes                     1142
```

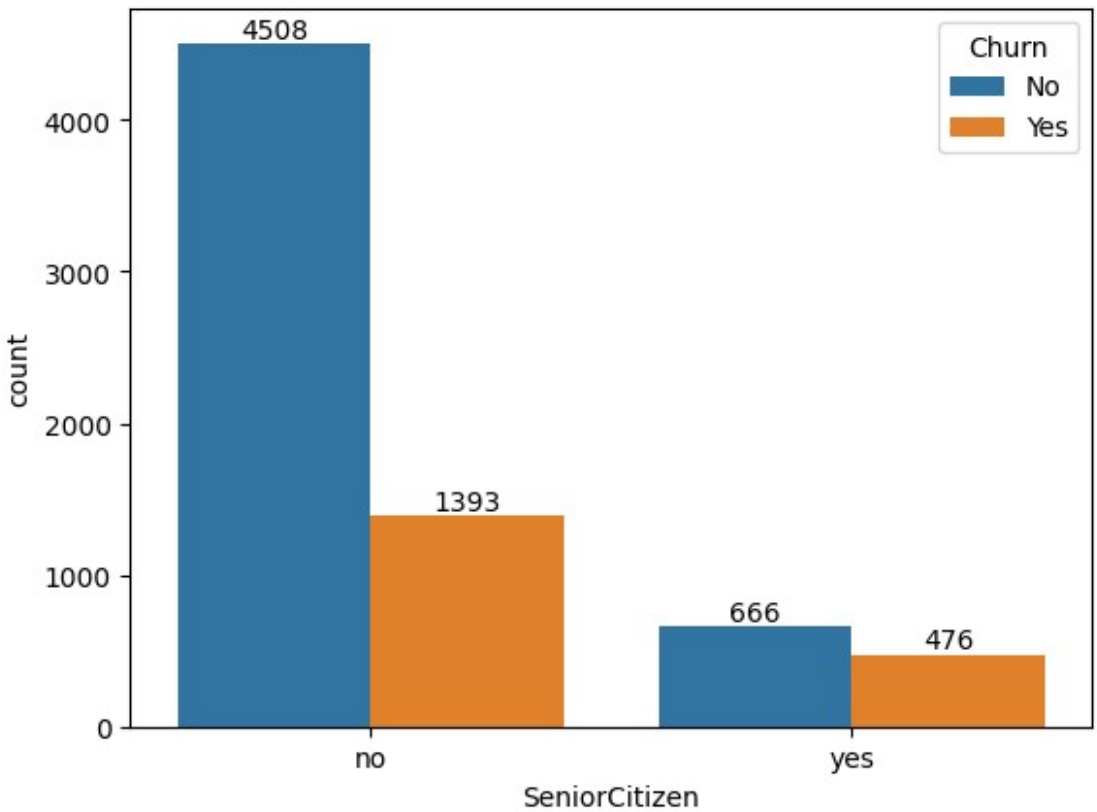
```
plt.pie(x=gb['SeniorCitizen'], labels=gb.index, autopct='%1.2f%%')
plt.title("Senior Citizen distribution")
plt.show()
```




```
df["SeniorCitizen"].value_counts()
```

```
SeniorCitizen
no     5901
yes    1142
Name: count, dtype: int64
```

```
ax=sns.countplot(data=df,x="SeniorCitizen",hue='Churn')
for i in ax.containers:
    ax.bar_label(i)
plt.show()
```

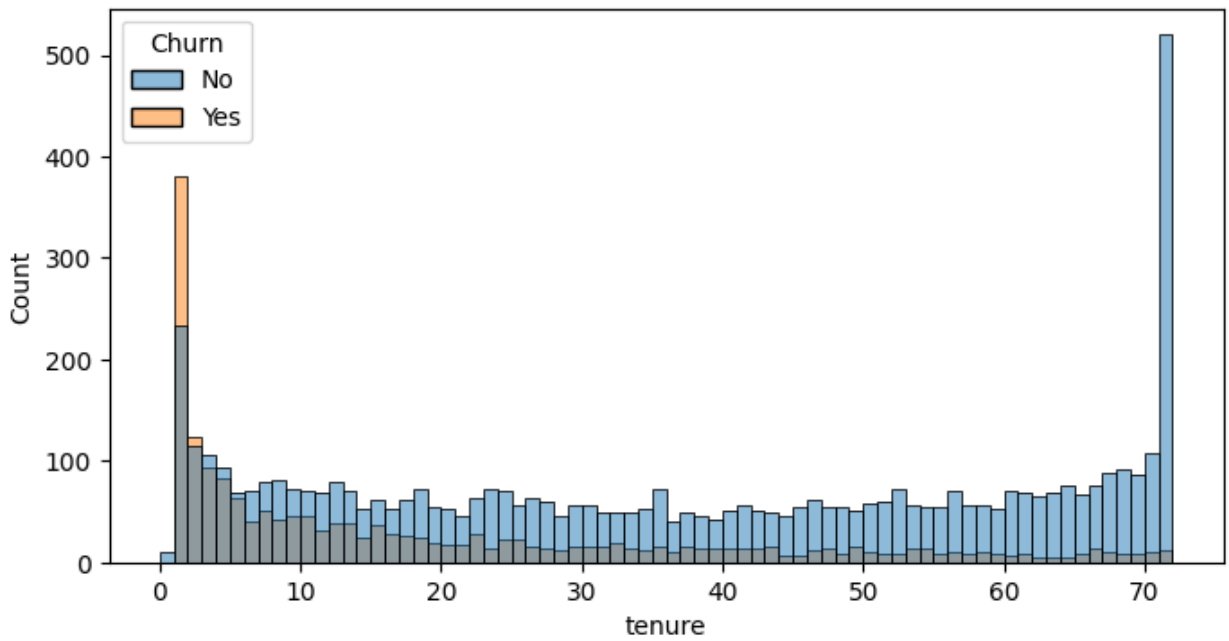



```
senior_churn_per=(476/1142)*100
non_senior_churn_per=(1393/5901)*100
print(f"Senior churn percentage",senior_churn_per,"%")
print(f"Non Senior churn percentage",non_senior_churn_per, "%")
```

```
Senior churn percentage 41.68126094570928 %
Non Senior churn percentage 23.606168446026096 %
```

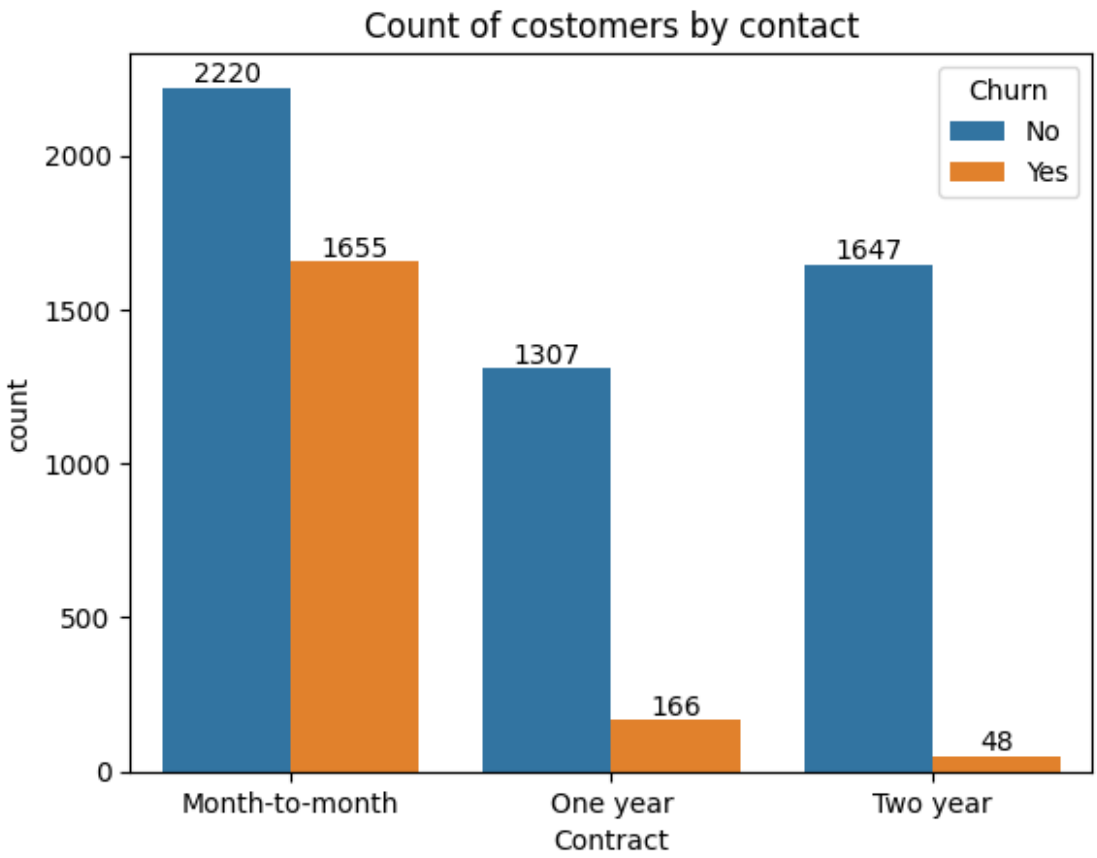
Comparatibly a greater number of Senior Citizen are churned out

```
plt.figure(figsize=(8,4))
sns.histplot(data=df, x="tenure", bins=72, hue='Churn')
plt.show()
```



people which are stay for long time has stayed and people whi are churned in the early stage

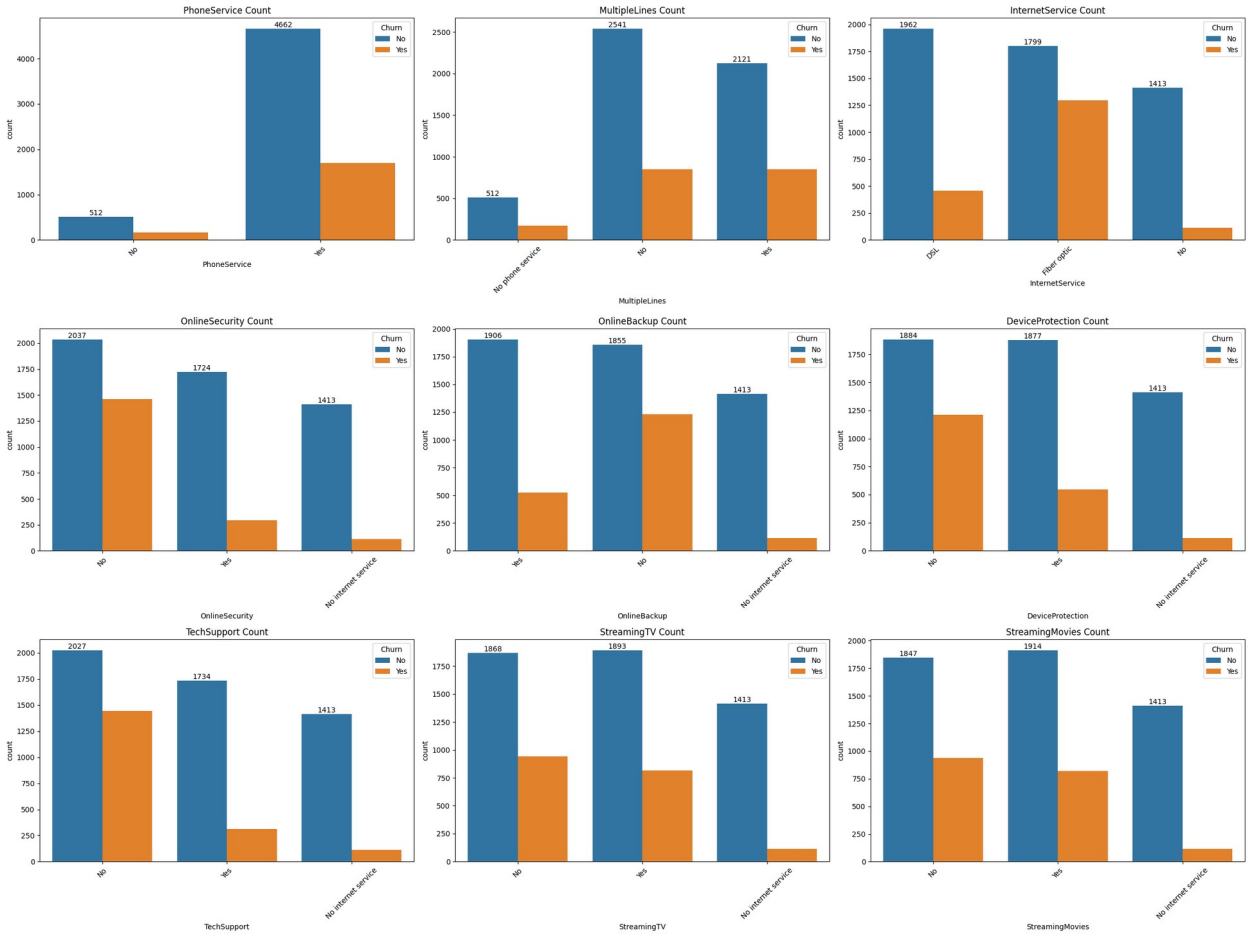
```
ax=sns.countplot(data=df, x="Contract", hue='Churn')
for i in ax.containers:
    ax.bar_label(i)
plt.title("Count of costomers by contact")
plt.show()
```



People who are month to month contract are likly to churn as compared to the 1 yr or 2 yr plane people

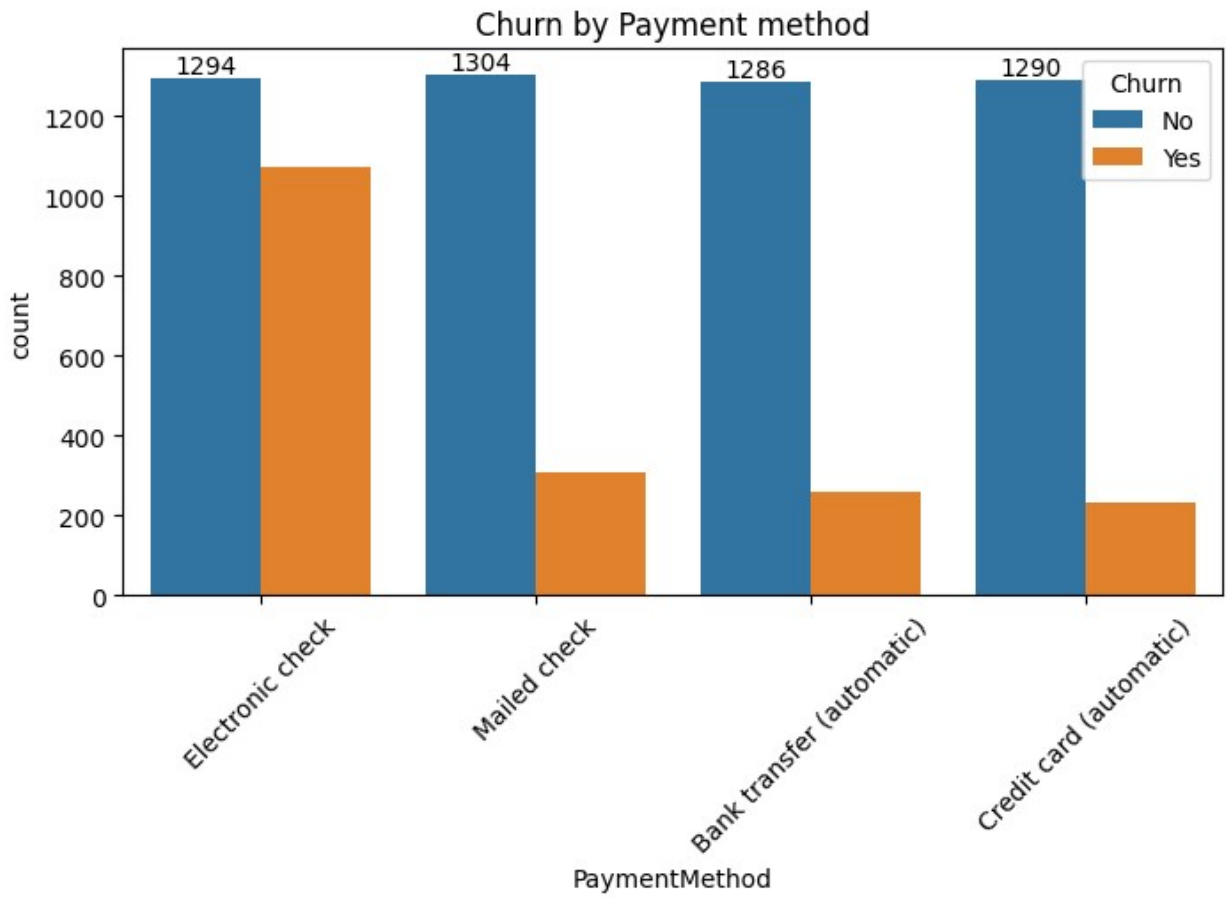
```
df.columns.values

<StringArray>
[      'customerID',           'gender',    'SeniorCitizen',
          'Partner',       'Dependents',           'tenure',
     'PhoneService',    'MultipleLines',  'InternetService',
   'OnlineSecurity',     'OnlineBackup', 'DeviceProtection',
      'TechSupport',      'StreamingTV',  'StreamingMovies',
         'Contract', 'PaperlessBilling',    'PaymentMethod',
   'MonthlyCharges',     'TotalCharges',            'Churn']
Length: 21, dtype: str

# your columns
cols = [
    'PhoneService', 'MultipleLines', 'InternetService',
    'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
    'TechSupport', 'StreamingTV', 'StreamingMovies'
]

# set subplot grid (3x3 since 9 columns)
fig, axes = plt.subplots(3, 3, figsize=(24,18))
```

```
axes = axes.flatten()  # flatten to iterate easily


# loop through columns and plot
for i, col in enumerate(cols):
    s=sns.countplot(x=col, data=df, ax=axes[i], hue="Churn")
    s.bar_label(s.containers[0])
    axes[i].set_title(f'{col} Count')
    axes[i].tick_params(axis='x', rotation=45)


# adjust layout
plt.tight_layout()
plt.savefig("subplot grid.png")
plt.show()
```




```
plt.figure(figsize=(8,4))
ax=sns.countplot(data=df, x='PaymentMethod', hue='Churn')
ax.bar_label(ax.containers[0])
plt.title("Churn by Payment method")
```

```
plt.xticks(rotation=45)
plt.show()
```



Costomer is likely to churn when he using Electronic check Payment method